Engineering Problem Framing Challenge

Blanket / Worksheet Template for Group Work (30–40 minutes)

Team members: _____ **Scenario title:** _____

Instructions: Read your scenario carefully. Use evidence from the text to classify statements, frame the problem, draw a process flow diagram, define variables with units, map stakeholder views, and propose must-have and nice-to-have targets. Keep the problem statement solution-free.

A. Sort the Evidence

Type	Statement from scenario	Why this category?
Fact #1		
Fact #2		
Inference #1		
Inference #2		
Speculation #1		
Speculation #2		
Opinion #1		
Opinion #2		

B. Complaint → Need → Problem Statement

Complaint (what people say):

_____
_____

_____
_____

Need (what they actually require):

_____
_____

_____
_____

Solution-free problem statement:

_____
_____

_____
_____

Template: [Stakeholder] needs [outcome] in [system], judged by [measure], within [constraints].

C. Process Flow Diagram (6–8 major steps)

Use: Start/End, Step, Decision (Yes/No), Wait/Queue, and Flow arrows. Label what moves between steps and note variables/units where possible.

D. Variables on the Process

List at least 5 variables. Include at least 1 performance variables, 1 design variable, and 1 context variable.

Step / Flow	Variable	Unit	Type (P / D / C)	How measured?

E. Stakeholder Performance View

Stakeholder	Their idea of success	Candidate variable	Unit

F. Must-have and Nice-to-have Targets

For at least 3 variables, propose rough targets. Must-have = demand. Nice-to-have = wish.

Variable	Current value (if known)	Unit	Must-have target	Nice-to-have target

G. Final Working Definition of System Performance

Complete the sentence: "The performance of _____ is judged by _____, _____, and _____, each with units and a target, as seen by _____."

_____
_____

_____
_____

H. 60-Second Team Report-Out (optional)

- What is your solution-free problem statement?
- What are the 2–3 most important variables and units?
- Which stakeholder matters most in your framing, and why?
- What is one must-have target and one nice-to-have target?